

How Secure is Online Banking?

In a survey conducted by the Online Banking Association, member institutions rated security as the most important issue of online banking. There is a dual requirement to protect customers' privacy and protect against fraud. The software generally used by banks for online banking is secure, tried and tested. Browsers offer from 40-bit to 128-bit encryption. Indian cyber laws too have given legitimacy to digital signatures.

In Netscape versions 3.x and earlier, you can tell what kind of encryption is in use for a particular document by looking at the 'document information' screen accessible from the file menu. The little key in the lower left-hand corner of the Netscape window also indicates this information. A solid key with three teeth means 128-bit encryption, a solid key with two teeth means 40-bit encryption, and a broken key means no encryption.

Even if your browser supports 128-bit encryption, it may use 40-bit encryption when talking to other servers or to servers outside the US and Canada. In Netscape versions 4.x and higher, click on the 'Security' button to

determine whether the current page is encrypted and, if so, what level of encryption is in use.

In the Microsoft Net Explorer, a solid padlock will appear on the bottom right of the screen when encryption is in use. To determine whether 40-bit or 128-bit encryption is in effect, open the document information page using **File > Properties**. This will indicate whether 'weak' or 'strong' encryption is in use.

SSL uses public-key encryption to exchange a session key between the client and server. This session key is used to encrypt the http transaction (both request and response). Each transaction uses a session key so that even if someone did manage to decrypt a transaction, they cannot find the server's secret key. If they wanted to decrypt another transaction, they'd need to spend as much time and effort on the second transaction as they did on the first.

Servers and browsers do encryption ranging from a 40-bit secret key to a 128-bit secret key, that is to say '2 to the 40th

power' or '2 to the 128th power'. In a 40-bit key, you will have to keep on trying each of the 2^{40} possible keys until you find the one that decrypts the message. This was in fact demonstrated when a French researcher used a network of fast workstations to crack a 40-bit encrypted message in a little over a week. An online credit card transaction is completed in a few moments, not giving enough time to crack your password.

Of course, using a 128-bit key makes it more difficult because there are 2^{128} instead of 2^{40} possible keys. Using the same method (a network of fast workstations) to crack a message encrypted with such a key would take significantly longer than the age of the universe using conventional technology. 128-bit is not just 'three times' as powerful as 40-bit encryption. 2^{128} is 'two times two, times two, times two...' with 128 two's. That is two, doubled on itself 128 times. 2^{40} is already a huge number, about a trillion (that's a million, million!). Therefore, 2^{128} is that number (a trillion), doubled over and over on itself another 88 times.

checks the funds/shares availability and automatically credits/debits the accounts once the order is executed by the exchange.

A trader can trade shares on margin at rates as low as 0.10 per cent. It's important to compare various online trading sites on brokerage rates, inclusive of all sub-charges. The tickers available on these sites provide instantaneous updates. Also, some Web sites can offer to transact in those shares instantaneously and with convenience.

Online trade confirmations reach ICICI customers within 4 minutes, while contract notes are dispatched at the end of the day and reach within 24 to 36 hours.

Investors trading online can also do their technical analysis, know what other brokers are recommending—buy or a sell,

access live news from international agencies such as Reuters and CNBC, and find out what the leading CEOs think about the state of the economy and the capital market.

Are you buying yet?

In India, while the population runs into billions, Internet surfers amount to a very miniscule percentage. An IDC report states that there are almost a million active Net banking users—that might just be 0.01 per cent of the total population, but they represent 15 per cent of the Net user populace.

One expects banks to woo customers to logon and mutually save time and money. In the US, for instance, banks encourage customers to transact online by crediting around Rs 1,250 to them. Paying you money to transact online is a remote thought in India. Just look at the terms of agreement (the small print as usual). They generally read: "...the user hereby fully indemnifies and holds the (Bank) and its affiliates harmless against any action, suit, proceeding initiated against it or any loss, cost or damage incurred by it as a result thereof..."

You will notice that any possible risk (see box, 'How Secure is Online

banking?') connected to online banking is passed on to you. If a hacker manages to crack the password or commit any breach, the customer has to pay even if he was not at fault. In the US, it's not so. Banks cover the risk of hacking and the customer is encouraged to perform online financial transactions.

There are quite a few services missing in the best of online banks in India. They include Online PPF accounts. Many nationalised banks have PPF accounts. If only these were made available online. One would also love to pay taxes online.

Fully functional online banking will take a long time to become as commonplace as automated teller machines (ATMs). And then, virtual banks can't give you cash for which you have to go to an ATM. Also, some banking Web sites can be difficult to navigate at first. Besides, a lot of fine print needs to be checked.

However, for many people, the biggest hurdle to online banking is learning to trust it. Did my transaction go through? Did I push the transfer button once or twice? While the government, banks and financial institutions can do their bit in educating the public, on our part, we need to take a first step—that is, get an Internet banking facility. ■



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